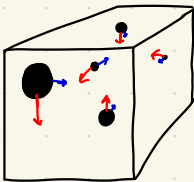


Goal: Plot an assortment of orbital bodies with randomly generated Velocity vectors in a bounded box. Go through each body and calculate the vector of gravitational force on it. Then add a visible red arrow for the velocity and a blue one for gravity.

e.g.



System Defaults

Unit of Mass = 1 Solar mass (SunMass)

Unit of distance = 1 Astronomical unit

Unit of time = 1 day

Unit of energy = $\frac{(\text{Solar mass}) \cdot (\text{AU})^2}{(\text{Day})^2}$

Gravitational Constant = 0.000295913120346 (See 1.1)

New Gravitational Constant 1.1

Current $G = 6.6743 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$
 New $G = 6.6743 \times 10^{-11} \frac{(\text{Solar mass}) (\text{day})^2}{(\text{AU})^3}$

- AU = 149,597,870,700 meters

- Solar Mass = 1.98846×10^{30} Kg

- Day = 86,400 seconds

New $G = 0.000295913120346$

Equation 1.2

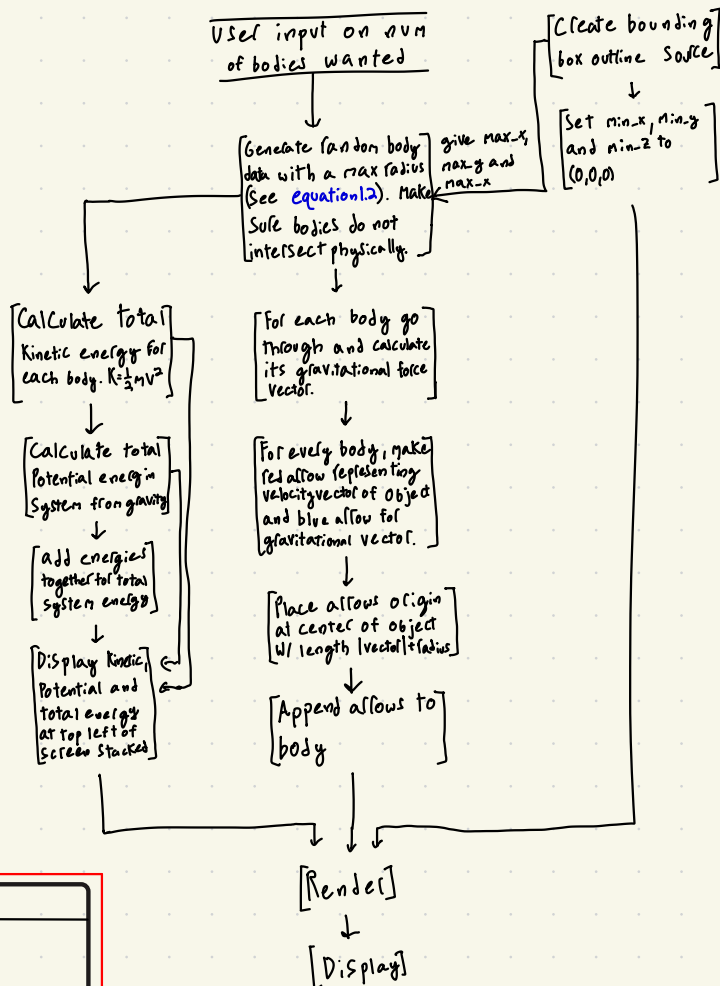
X # of bodies with variable radius r in a afa. Let q be a random ratio that determines max density of bodies wanted in System. The max variable size of r ($\text{max}(r)$) is determined by equation: $\text{Max}(r) = q \cdot \frac{a}{2x}$

Custom Classes

Body
~Position Vector
~radius: Double
+ distance Squared (Body): Double
+ distanceTo (Body): Double
+ distanceTo (Body): Bool
+ isIn Bounds(): Bool

NewtonianBody
- Velocity Vector
- Gravity Vector
- mass: Double
+ GRAVITATIONAL_CONSTANT
+ setGravity(bodies[]): void
+ getMyKinetic(): Double
+ getMyPotential(): Double
+ getSystemKinetic(bodies[]): Double
+ getSystemPotential(bodies[]): Double

Functional Diagram



Object Diagram

[# of bodies]



[NewtonianBody[]]
list of randomly generated
NewtonianBody objects
w/ max possible radius
following equation 1.1



[VTK DoubleArray
Array of gravity
vector for each body]

[VTK DoubleArray
Array of velocity
vector for each body]

[VTK DoubleArray
Array of radius
for each body]

[VTK Points
list of position
vector for
each body]

[VTK Arrow Source
Velocity vector]

[VTK Arrow Source
Gravity vector]

[VTK SphereSource
basis for each body]



[VTK Poly Data]
Stores the following:
Points: positioning
Array: radii
Array: Velocity vect.
Array: Gravity vect.

[VTK TextActor
Total kinetic
energy]

[VTK TextActor
Total system
energy]

[VTK TextActor
Total potential
energy]

[VTK CubeSource]
Cube covering the
bounding box area

[VTK Mapper]

[VTK Actor]
Set representation
of property to
wire frame

[VTK Glyph3D]
Places velocity
arrows on screen
at their point

[VTK Mapper
maps vel. glyph]

[VTK Actor]

[VTK Glyph3D]
Places gravity
arrows on screen
at their point

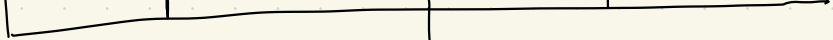
[VTK Mapper
Maps grav. glyph]

[VTK Actor]

[VTK Glyph3D]
Places spheres at
their specified
coords and with
their radius

[VTK Mapper
Maps body glyph]

[VTK Actor]



[Renderer]

↓
[Window]